

Exact Enumeration and Scalar Recurrences for Left-Right Orbits of Subsets of Permutations

Candidate full solution for arXiv:2503.09542

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Status: candidate full solution, likely valid. The formula below determines every $f_{n,k}$ in Question 3.5 of Kushwaha–Tripathi [1]. A constructive Wronskian elimination then gives a scalar finite-shift recurrence in k for every fixed n .

1 Source question

Kushwaha and Tripathi define $\mathcal{P}_{n,k} = \{A \subseteq S_n : |A| = k\}$ and declare $A \sim B$ if $B = \mu A \nu$ for some $\mu, \nu \in S_n$. Their Question 3.5 (PDF page 12, source label `ques:Equivalence`) is:

Determine the number of equivalence classes in $\mathcal{P}_{n,k}$ under $\sim$, that is, determine $f_{n,k} = |\mathcal{P}_{n,k}/\sim|$. Does $f_{n,k}$ satisfy some recurrence?

Number of orbits of subsets of permutations. We finally bring the reader's attention to another curious phenomenon in Table 1. Looking at the last two columns in Table 1 shows that Erdős matrices typically have large equivalence classes. Recall that we say two bistochastic matrices A and B are equivalent, $A \sim B$, if $A = PBQ$ for some permutation matrices P, Q . Given a bistochastic matrix A , can we understand the size of the equivalence class of A ? This leads us to propose the following combinatorics problem that may be of independent interest.

Let $n \in \mathbb{N}$ and let $k \in [n!]$. Let $\mathcal{P}_{n,k}$ be the family of all subsets of size k of the permutations in S_n . That is,

$$\mathcal{P}_{n,k} := \{S \subseteq S_n : |S| = k\}.$$

Given $S = \{\sigma_1, \dots, \sigma_k\}, T = \{\tau_1, \dots, \tau_k\} \in \mathcal{P}_{n,k}$ we declare $S \sim T$ if there exist permutations $\mu, \nu \in S_n$ such that

$$T = \mu S \nu := \{\mu \sigma_1 \nu, \dots, \mu \sigma_k \nu\}.$$

It is easy to check that $\sim$ is an equivalence relation on $\mathcal{P}_{n,k}$.

Question 3.5. *Determine the number of equivalence classes in $\mathcal{P}_{n,k}$ under $\sim$. That is, determine $f_{n,k} := |\mathcal{P}_{n,k}/\sim|$. Does $f_{n,k}$ satisfy some recurrence?*

It is easy to note that $f_{n,1} = f_{n,n!} = 1$ for any n . It is also easy to show that $f_{n,2} = p(n) - 1$ for all $n \geq 2$.

Figure 1: The definition and Question 3.5 in the source paper, PDF page 12.

2 Notation and explicit formula

Write a partition $\lambda \vdash n$ as $\lambda = (1^{m_1} 2^{m_2} \dots)$ and put

$$z_\lambda = \prod_{j \geq 1} j^{m_j} m_j!.$$

Thus z_λ is the centralizer size of a permutation of type λ . For $r \geq 1$, let $\lambda^{[r]}$ be the cycle type of the r -th power of such a permutation. Explicitly, every part j of λ is replaced by $\gcd(j, r)$ parts of size $j/\gcd(j, r)$. Also set

$$o(\lambda) = \text{lcm}\{j : m_j > 0\}.$$

For $\lambda, \eta \vdash n$, let $L_{\lambda, \eta} = \text{lcm}(o(\lambda), o(\eta))$. For every $r \mid L_{\lambda, \eta}$ define

$$a_r(\lambda, \eta) = \begin{cases} z_{\lambda^{[r]}}, & \lambda^{[r]} = \eta^{[r]}, \\ 0, & \lambda^{[r]} \neq \eta^{[r]}, \end{cases}$$

and, for every $\ell \mid L_{\lambda, \eta}$, define

$$c_\ell(\lambda, \eta) = \frac{1}{\ell} \sum_{r \mid \ell} \mu_M(\ell/r) a_r(\lambda, \eta),$$

where μ_M is the number-theoretic Moebius function.

Theorem 1 (Exact orbit formula). *For every $n \geq 1$ and $0 \leq k \leq n!$,*

$$f_{n,k} = \sum_{\lambda, \eta \vdash n} \frac{1}{z_\lambda z_\eta} [x^k] \prod_{\ell \mid L_{\lambda, \eta}} (1 + x^\ell)^{c_\ell(\lambda, \eta)}. \quad (1)$$

In particular, the right-hand side is a finite explicit determination of $f_{n,k}$ in terms of partitions of n .

3 Proof intuition

Replace the paper's right multiplier by its inverse, which does not change the equivalence relation. A pair (μ, ν) then permutes S_n by $\sigma \mapsto \mu\sigma\nu^{-1}$. A subset is fixed precisely when it is a union of cycles of this permutation, so its size-generating polynomial is a product of factors $1 + x^{\text{cycle length}}$.

The only non-obvious input is that all of those cycle lengths can be recovered without iterating through S_n . The fixed points of the r -th power are exactly the conjugators between ν^r and μ^r . Their number is either zero or the centralizer size of their common power type. Moebius inversion turns these fixed-point counts into exact action-cycle counts. Burnside's lemma then averages the fixed-subset polynomial, and grouping the average by the two conjugacy types gives (1).

For the recurrence clause, regard the finitely many fixed-subset polynomials as an explicitly known polynomial space. A basis and its Wronskian produce a scalar differential equation annihilating every polynomial in that space, including $F_n(x) = \sum_k f_{n,k} x^k$. Coefficient extraction gives a recurrence for $f_{n,k}$ whose coefficients are computed only from the partition data, not from a previously expanded row of orbit counts.

4 Proof

For $\mu, \nu \in S_n$, define a permutation of the set S_n by

$$T_{\mu, \nu}(\sigma) = \mu\sigma\nu^{-1}.$$

These transformations give the usual $S_n \times S_n$ action, and their orbits on k -subsets are exactly the equivalence classes in the question.

Suppose μ and ν have cycle types λ and η . Since

$$T_{\mu,\nu}^r(\sigma) = \mu^r \sigma \nu^{-r},$$

we have

$$\begin{aligned} \sigma \in \text{Fix}(T_{\mu,\nu}^r) &\iff \mu^r \sigma \nu^{-r} = \sigma \\ &\iff \mu^r = \sigma \nu^r \sigma^{-1}. \end{aligned}$$

Thus there are no fixed points unless $\lambda^{[r]} = \eta^{[r]}$. If the two power types agree, the set of conjugators is a coset of the centralizer of ν^r , and hence

$$|\text{Fix}(T_{\mu,\nu}^r)| = a_r(\lambda, \eta). \quad (2)$$

Let C_ℓ denote the number of ℓ -cycles of $T_{\mu,\nu}$. Its order divides $L_{\lambda,\eta}$. For each $r \mid L_{\lambda,\eta}$, a cycle of length ℓ contributes all of its ℓ points to $\text{Fix}(T^r)$ exactly when $\ell \mid r$. Therefore

$$a_r(\lambda, \eta) = \sum_{\ell \mid r} \ell C_\ell.$$

Moebius inversion gives

$$C_\ell = \frac{1}{\ell} \sum_{r \mid \ell} \mu_M(\ell/r) a_r(\lambda, \eta) = c_\ell(\lambda, \eta).$$

This also proves directly that every $c_\ell(\lambda, \eta)$ defined above is a nonnegative integer.

A subset of S_n is fixed by $T_{\mu,\nu}$ if and only if it is a union of cycles of $T_{\mu,\nu}$. Consequently, the number of its fixed subsets of size k is

$$[x^k] \prod_{\ell \mid L_{\lambda,\eta}} (1 + x^\ell)^{c_\ell(\lambda,\eta)}. \quad (3)$$

Burnside's lemma [2] averages (3) over the $(n!)^2$ pairs (μ, ν) . There are $n!/z_\lambda$ permutations of type λ and $n!/z_\eta$ permutations of type η . Grouping the average by (λ, η) therefore replaces the multiplicity divided by $(n!)^2$ by $1/(z_\lambda z_\eta)$, which yields (1). $\square$

5 The scalar recurrence

Equation (1) gives the whole row generating polynomial

$$F_n(x) = \sum_{k=0}^{n!} f_{n,k} x^k = \sum_{\lambda, \eta \vdash n} \frac{1}{z_\lambda z_\eta} \prod_{\ell \mid L_{\lambda,\eta}} (1 + x^\ell)^{c_\ell(\lambda,\eta)}.$$

It immediately provides an exact algorithm indexed by $p(n)^2$ pairs of partitions, rather than by the $\binom{n!}{k}$ subsets in the original definition. It also recovers the source's complement symmetry $f_{n,k} = f_{n,n!-k}$, because every factor in a fixed-subset polynomial is palindromic of total degree $n!$.

Set

$$Q_{\lambda,\eta}(x) = \prod_{\ell \mid L_{\lambda,\eta}} (1 + x^\ell)^{c_\ell(\lambda,\eta)}$$

and let V_n be the rational linear span of all these polynomials. Since every $Q_{\lambda,\eta}$ is given explicitly by cycle-type data, a basis $Q_1, \dots, Q_{m_n}$ of V_n is obtained constructively by row reduction of their coefficient vectors.

For $0 \leq j \leq m_n$, define

$$A_{n,j}(x) = (-1)^{m_n+j} \det \left(Q_i^{(r)}(x) \right)_{\substack{0 \leq r \leq m_n, \\ 1 \leq i \leq m_n, \ r \neq j}}. \quad (4)$$

In particular, A_{n,m_n} is the Wronskian of the chosen basis and is nonzero over the characteristic-zero field $\mathbb{Q}$ [3].

Theorem 2 (Constructive scalar recurrence). *For every fixed n , the row generating polynomial satisfies*

$$\sum_{j=0}^{m_n} A_{n,j}(x) F_n^{(j)}(x) = 0. \quad (5)$$

Write $A_{n,j}(x) = \sum_t a_{n,j,t} x^t$, extend $f_{n,q} = 0$ outside $0 \leq q \leq n!$, and write $q^{\underline{j}} = q(q-1) \cdots (q-j+1)$. Then, for every $r \geq 0$,

$$\boxed{\sum_{j=0}^{m_n} \sum_t a_{n,j,t} (r+j-t)^{\underline{j}} f_{n,r+j-t} = 0.} \quad (6)$$

The polynomials $A_{n,j}$, hence all coefficients in this scalar recurrence, are computed from the partition power-types in the orbit formula without using the values of $f_{n,k}$.

Proof. Append a last column consisting of $y, y', \dots, y^{(m_n)}$ to the derivative matrix of $Q_1, \dots, Q_{m_n}$. Expanding its determinant along that last column gives the left side of (5). If $y \in V_n$, the columns are linearly dependent, so the determinant vanishes. Formula (1) places F_n in V_n , proving (5). The coefficient of x^r in $x^t F_n^{(j)}(x)$ is $(r+j-t)^{\underline{j}} f_{n,r+j-t}$. Extracting $[x^r]$ proves (6). $\square$

There is also a lower-dimensional practical recurrence before scalar elimination. If $q_{\lambda,\eta,k} = [x^k]Q_{\lambda,\eta}(x)$, put

$$b_{\lambda,\eta,s} = \sum_{\ell|s} (-1)^{s/\ell-1} \ell c_\ell(\lambda, \eta).$$

The logarithmic derivative of $Q_{\lambda,\eta}$ gives

$$q_{\lambda,\eta,0} = 1, \quad k q_{\lambda,\eta,k} = \sum_{s=1}^k b_{\lambda,\eta,s} q_{\lambda,\eta,k-s}, \quad (7)$$

and the desired scalar count is the weighted sum $f_{n,k} = \sum_{\lambda, \eta \vdash n} (z_\lambda z_\eta)^{-1} q_{\lambda,\eta,k}$. Thus both the scalar existence statement and an efficient recursive computation are explicit.

Example: the complete recurrence for $n = 3$

For $n = 3$, the six distinct fixed-subset polynomials span a four-dimensional space. Wronskian elimination gives

$$\begin{aligned} 0 = & (x^2 - 1)^3 F_3^{(4)} - 12x(x^2 - 1)^2 F_3^{(3)} + 12(x^2 - 1)(6x^2 - 1) F_3'' \\ & - 120x(2x^2 - 1) F_3' + 360x^2 F_3. \end{aligned}$$

Equivalently, with $f_{3,q} = 0$ outside $0 \leq q \leq 6$, for every $r \geq 0$,

$$\begin{aligned} 0 = & -(r+1)(r+2)(r+3)(r+4)f_{3,r+4} \\ & + 3(r-4)(r-1)(r+1)(r+2)f_{3,r+2} \\ & - 3r(r-6)(r-5)(r-3)f_{3,r} \\ & + (r-8)(r-7)(r-6)(r-5)f_{3,r-2}. \end{aligned}$$

Together with $(f_{3,0}, f_{3,1}, f_{3,2}, f_{3,3}) = (1, 1, 2, 2)$, the cases $r = 0, 1, 2$ produce the rest of the row $(2, 1, 1)$.

6 Scope

Theorem 2 answers the recurrence clause in the natural coefficient direction: for every fixed n , $k \mapsto f_{n,k}$ satisfies a constructively specified scalar finite-shift recurrence. It does not assert a low-order recurrence uniform in n , nor a recurrence coupling different values of n . The source question asks only whether $f_{n,k}$ satisfies “some recurrence” and does not prescribe such stronger uniformity. The paper’s random linear-independence conjecture and Gram-matrix positivity question are also outside the scope of this result.

7 Exact computational verification

The included script `code/verifier.py` implements every term in (1) with integer and rational arithmetic. Run from the repository root:

```
conda run --no-capture-output -n sandbox python \
  runs/fa_banach_001/solutions/full/\
  2503.09542_orbit_count_cycle_index_formula/code/verifier.py
```

It reproduces every source-table value through $n = 5$, $k = 12$:

n	tested k	final tested value
2	$0 \leq k \leq 2$	$f_{2,2} = 1$
3	$0 \leq k \leq 6$	$f_{3,6} = 1$
4	$0 \leq k \leq 12$	$f_{4,12} = 5050$
5	$0 \leq k \leq 12$	$f_{5,12} = 732149722382$

As a second check, the script directly enumerates the cycles of $\sigma \mapsto \mu\sigma\nu$ for all 39 pairs of conjugacy types across $1 \leq n \leq 4$ and compares them to the power-type/Moebius formula. All comparisons agree. These tests are consistency evidence; the proof does not depend on them.

The second script `code/recurrence_verifier.py` computes the dimensions of the explicit fixed-subset polynomial spans for $1 \leq n \leq 5$:

n	1	2	3	4	5
m_n	1	2	4	10	19

It constructs the Wronskian operator for $n = 3$ and verifies both the displayed differential equation and every coefficient-recurrence residual exactly. Run it with

```
conda run --no-capture-output -n sandbox python \
  runs/fa_banach_001/solutions/full/\
  2503.09542_orbit_count_cycle_index_formula/code/recurrence_verifier.py
```

8 Novelty audit and review recommendation

The bounded search (performed 26 August 2026) covered the run registry and solution indexes; arXiv id 2503.09542; the exact title and question wording; searches for left-right subset orbits together with Burnside, cycle-index, Wronskian, and recurrence keywords; and OEIS A381842 [4]. The source contains no occurrence of “Burnside”, “Polya”, or “cycle index”. The OEIS entry, last modified 16 July 2026, lists only $f_{n,2} = p(n) - 1$, complement symmetry, and initial values. A fresh recurrence-focused search on 26 August 2026 found no later paper or matching formula.

Novelty confidence is nevertheless only *moderate*: this is a natural finite-group enumeration argument and could appear in older literature under different terminology such as two-sided translations of subsets. Human review should first verify equations (2), (3), and the Wronskian elimination in Theorem 2, then commission a specialist historical search before any public novelty claim. The mathematical validity of both the formula and recurrence appears high. A reviewer should also confirm that recurrence in k is the intended reading of the source’s deliberately unspecified recurrence clause.

References

- [1] A. Kushwaha and R. Tripathi, *A note on Erdős matrices and Marcus–Ree inequality*, Linear Algebra Appl. **725** (2025), 223–247; [arXiv:2503.09542](#).
- [2] P. J. Cameron, *Combinatorics: Topics, Techniques, Algorithms*, Cambridge University Press, 1994.
- [3] E. L. Ince, *Ordinary Differential Equations*, Dover Publications, 1956.
- [4] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, [A381842](#), accessed 26 August 2026.